Data Quality Plan

Introduction

The Data Quality Report provided an initial exploration of the training dataset and identified several structural and statistical issues affecting the Analytics Base Table (ABT). While basic cleaning steps were applied, a number of data quality challenges remain. These include missing values, high-cardinality features, inconsistent categorical labels, and the presence of extreme values with the target variable, price.

The purpose of the Data Quality Plan is to formally address these issues by evaluating potential handling strategies and selecting appropriate preprocessing steps for each problematic feature. As outlined by Kelleher, Mac Namee and D'Arcy (2020), once the structure and properties of the data are understood, the next stage is to identify data quality issues and determine how they should be handled before predictive modelling can take place.

This section outlines the identified issues, discusses possible solutions, and specifies the final preprocessing decisions and feature types that will be used in the cleaned dataset.

Summary

The table below summarises the key features identified as having potential data quality issues, along with possible strategies for addressing them. These strategies were evaluated before selecting a final preprocessing approach for each feature.

Feature	**Data Quality Issue**	**Potential Handling Strategies**
Price	Extreme outliers/ heavy right skew	Remove outliers, clamp values
Eircode	Missing values (~69%), extremely high cardinality	Remove feature, extract routing key
Description of Property	Irish/English translation consistency	Convert Irish to English
Property Size Description	Very high missingness (~95%)	Impute, treat missing as a category, remove feature
Address	High cardinality text, formatting inconsistent, mixed commas/spacing	Remove feature, standardise formatting
Date of Sale	None	Addressed in feature engineering (part 4)
VAT Exclusive	High Overlap with Description of Property (99.7% overlap)	Keep in cleaned data, will be excluded in final model
County	None	N/A
Not Full Market Price	None	N/A

Selected Handling Strategies and Final Feature Types

Price

The Price feature exhibited substantial variability and extreme values, as identified in the DQR through descriptive statistics and visualisations such as boxplots and histograms. The distribution of property prices was heavily right-skewed, with a small number of extremely high-value transactions producing a long upper tail.

Several strategies were considered for handling these extreme values. One option was to remove observations classified as outliers. However, this approach is typically used when outliers are believed to result from data entry errors (Kelleher et al.). In this dataset, the extreme values seem to represent legitimate high-value property transactions, therefore removing them would unnecessarily reduce the dataset size.

Another option was to apply a clamping approach. Initially, the interquartile range method described by Kelleher et al. (2020) was considered, where the lower bound is defined as the first quartile minus 1.5 times the IQR and the upper bound as the third quartile plus 1.5 times the IQR. However, applying this threshold resulted in clamping values that still appeared to be realistic property prices. For example, the lower bound of the upper clamped range was approximately €675,000, suggesting that many valid transactions would be affected.

Alternative percentile-based thresholds were also evaluated. A 99.5th percentile threshold was found to be substantially higher than the 99th percentile, resulting in insufficient mitigation of extreme values. The 99th percentile provided a more appropriate balance between reducing the influence of extreme values and preserving valid observations. Therefore, the Price feature was clamped at the 99th percentile. A lower bound clamp was also considered but ultimately not applied, as lower-price transactions appear to fall within a reasonable range.

Final feature type: **Continuous**

Eircode

The Eircode feature presents two significant data quality challenges: a high proportion of missing values and extremely high cardinality. Approximately 68.7% of observations do not contain an Eircode value, and among the non-missing entries there are over 16,000 distinct codes. Additionally, the most frequent Eircode occurs only a small number of times, indicating a very sparse distribution.

Several possible strategies were considered, such as extracting the routing key (first three characters of the Eircode) to reduce cardinality. However, according to Kelleher et al. (2020), a useful rule of thumb when dealing with missing values is that if the proportion of missing data in a feature exceeds 60%, the amount of information contained in that feature is likely too limited to be useful for predictive modelling. Since the Eircode feature exceeds this threshold, and we have alternative geographic indicators such as County and Address, the decision was made to remove the Eircode feature from the dataset at this stage.

More detailed geographic features will be derived from the Address variable in later stages of the project.

Description of Property

The DoP feature contains a small number of categorical levels describing whether a property is new or previously owned. During the data quality assessment, it was observed that the category "Second-Hand Dwelling house/ Apartment" also appeared in its Irish-language equivalent, "Teach/Árasán Cónaithe Athláimhe" (although it's slightly misspelled, missing the 'l' in the final

word). Although both labels refer to the same property type, they were recorded as separate categories in the dataset.

This inconsistency artificially increases the number of categorical levels and may reduce the effectiveness of the feature during modelling. To address this issue, the Irish translation was standardised by converting it to the corresponding English label.

Final feature type: **Categorical**

Property Size Description

The Property Size Description feature also exhibits a very high proportion of missing values, with approximately 94.8% of observations lacking size information. While several size categories are present among the non-missing observations, the extremely small proportion of available data limits the reliability of this feature.

As for the Eircode feature, Kelleher et al. note that when the proportion of missing values in a feature exceeds approximately 60%, the information contained in that feature is generally too limited to justify its inclusion in the Analytics Base Table. The PSD feature exceeds this threshold by a substantial margin, indicating that the feature provides very little usable information for predictive modelling.

Alternative strategies such as imputing missing values or treating missing entries as a separate category were regarded. However, given that the overwhelming majority of observations lack size information, any imputation would be largely speculative and could introduce misleading patterns into the dataset. Consequently, the decision was made to remove this feature entirely from the dataset.

Address

The Address feature contains free-form text with high cardinality but valuable geographic information. As it is not directly suitable for modelling, it was standardised to improve geocoding accuracy.

A custom cleaning function was implemented with the following key steps:

- Conversion to uppercase and trimming of whitespace
- Removal of unit-level prefixes (e.g. *APT, FLAT, UNIT, NO*) to focus on building-level addresses
- Collapsing of numeric ranges (e.g. *1–5*, *1 & 2*) into a single representative number
- Reduction of multiple leading numbers to a single number (e.g. *1 2 3 Street → 1 Street*)
- Normalisation of spacing and punctuation
- Appending standardised county format (*CO. [COUNTY]*) where missing

These steps were designed specifically to align addresses with Mapbox geocoding requirements, which perform best with consistent, building-level inputs.

Following standardisation, duplicate detection improved, indicating that formatting inconsistencies had previously masked identical addresses.

The cleaned address is not used directly as a model feature but is used to derive geographic coordinates (latitude and longitude), which are subsequently used for spatial enrichment.

Final feature type: **Text** (used for feature engineering only)

Date of Sale
The Date of Sale feature represents the date on which each property transaction occurred. During the initial cleaning phase, this feature was converted from string format to a datetime data type.

No further transformation was applied at this stage. While temporal features such as year or month may be useful for predictive modelling, these will be derived later during the feature engineering stage of the project. Retaining the full datetime format at this stage preserves all temporal information for relationship analysis in the next section.

Final feature type: **Datetime / Temporal**

Other Features
The remaining features in the dataset, including County, Not Full Market Price, and VAT Exclusive, were assessed and found to contain no significant data quality issues. These features contain no missing values and exhibit consistent formatting.

As a result, no preprocessing was required, and these features were retained in their original form.

Final feature types:
- County: Categorical
- Not Full Market Price: Binary categorical
- VAT Exclusive: Binary categorical

Column Standardisation
To improve usability and consistency across the pipeline, column names were standardised using lowercase and snake_case format. For example:

- *Price (€)* → price
- *Date of Sale (dd/mm/yyyy)* → date_of_sale
- *Description of Property* → property_description

This simplifies downstream processing and reduces the risk of errors when referencing columns throughout the analysis.

Reproducible Cleaning Function
To ensure consistency between training and test data, all preprocessing steps were encapsulated within a reusable function.

This function performs:

- Price cleaning and numeric conversion
- Application of the same clamping threshold derived from the training data
- Date conversion to datetime format
- Standardisation of categorical labels
- Conversion of categorical features to appropriate types
- Removal of duplicate rows
- Dropping of low-quality features (Eircode, Property Size Description)
- Column renaming for consistency

A key design decision was to apply the clamping threshold learned from the training dataset to the test dataset, rather than recalculating it. This prevents data leakage and ensures that model inputs are processed consistently.

Next Steps

The cleaned dataset produced in this stage contains no missing values and consistent feature formatting. In the next stage of the project, relationships between features will be explored, followed by the creation of new features to improve predictive performance.

Reference

Kelleher, J.D., Mac Namee, B. and D'Arcy, A. (2020) *Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies*. 2nd edn.